

MRI Scanner Synchronisation

About

There are several methods that can be used to synchronise functional MRI sequences to stimulus presentation. The best method to use will depend on the sequence type and stimulus presentation method.

MRI is treated differently to both [EEG and MEG synchronisation](#).

MRI Trigger Options

There are two types of 'MRI Trigger'.

- **Keypress Trigger**
 - **Volume:** a keypress '5' is output at the beginning of each acquisition volume.
 - **Slice:** a keypress '5' is output at the beginning of each acquisition slice.
- **Serial Trigger**
 - An output of serial data containing information for each acquisition slice (e.g Version, Reference, Filler, Slice).

The available trigger methods will depend on the MRI sequence and the stimulus package in use. Please use the following table to decipher the MR trigger options available to you.

	Cogent	Psychtoolbox	Psychopy
FIL 2D sequence	Keypress Serial	Keypress Serial	Keypress Serial (<i>No code currently available</i>)
FIL 3D sequence	Keypress Serial	Keypress Serial	Keypress Serial (<i>No code currently available</i>)
CMMR MultiBand sequence	Keypress	Keypress	Keypress
Using personal STIM-PC *Serial to USB convertor required	Keypress Serial*	Keypress Serial* (download code here: svn://bread/tools/scanner_triggers/trunk)	Keypress Serial* (<i>No code currently available</i>)

MRI Keypress Triggers

In MRI the triggers are sent: MRI scanner → Stimulus PC.

When using keypress triggers: a keypress of '5' (5% key) will be sent from the MRI scanner to the STIM-PC (though the keypad interface unit).

A keypress '5' can be sent at the start of each image **volume** or **slice**. This trigger type is a sequence parameter set by the Imaging Support or Physics team.

For unaccelerated protocols

e.g. the routine sequences described here:

http://intranet.fil.ion.ucl.ac.uk/wiki/physicswiki/doku.php?id=start:routine_sequences_prisma

Standard FIL advice is to acquire 5 imaging volumes before starting stimulus presentation.

This ensures an equilibrium longitudinal magnetisation is reached prior to scanning.

These 5 volumes will act as dummy scans and should be manually discarded from your data.

i.e. stimulus presentation lasting 200 volumes.

- Start script, waiting for keypress '5'
- Start MRI acquisition
- Count the incoming keypress '5' volume triggers
- Present stimulus after 5th keypress '5' received
- Total data 205 volumes
- Volumes 1-5 will be dummy scans to discard

For accelerated protocols

The first keypress five is sent on the first fMRI volume (after dummy, reference scans, and blip-reversed scans). The script should trigger on the first keypress five that the scanner sends and no data needs to be discarded. If you are running 3D EPI at 7T, and are using blip-reversed scans for top-up, those scans need to be separated from the rest of the time series prior to pre-processing and analysis. i.e. stimulus presentation lasting 200 volumes.

- Start script, waiting for keypress '5'
- Start MRI acquisition
- Present stimulus after 1st keypress '5' received
- Total data 200 volumes
- There will be no dummy scans to discard

Please contact fil.physics@ucl.ac.uk if you are unsure if your protocol is accelerated or has blip-reversed scans.

User Guide - Keypress Triggers

Please see a full user guide for setting up [stimulus presentation here](#).

1. Prior to scanning
2. Plug BNC cable '**Keypress 5**' into the 'Keypress

Trigger Unit'.

3. Ensure the 'Master USB' cable is connected to the STIM-PC in use.

Note: Failure to ensure both of these connections, will result in no keypress triggers being received.



MRI Serial Triggers

In MRI triggers are sent: MRI scanner → STIM-PC.

FIL MRI scanners send each slice number to the STIM-PC serial port 1 (COM 1). Slice numbers start from 1 and *do not* reset at the end of each volume. To use these slice numbers you will need to configure the serial port in your script, and use either the `getslice()` or `waitslice()` commands.

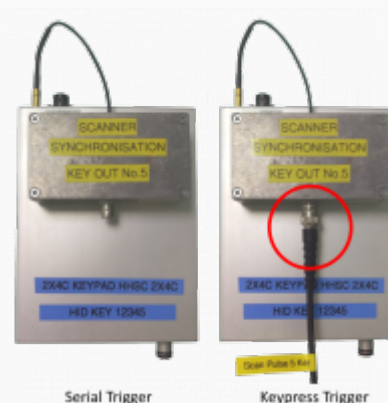
NB Following upgrades to 3T PRISMA and 7T MRI systems, a new parameter is required in `config_serial()` ; see example below!

User Guide - Serial Triggers

Please see a full user guide for setting up [stimulus presentation here](#).

1. Prior to scanning
2. Unplug BNC cable 'Scan Pulse 5' from the 'Keypress Trigger Unit'
3. Serial cables are attached to the WIN7 and WIN10 STIM-PC by default

Note: Failure to unplug this BNC cable will result in unexpected keypress outputs, likely into an open script.



Code Help - Serial Triggers

Typically one would wait for a particular slice number for each trial, or each block. For example:

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
port          = 1;                % scanners are connected to port
1.
slicespervol = 48;                % get this from your MRI
sequence
dummyvolumes = 6;                % typical value
dummyslices  = slicespervol * dummyvolumes;
slicelist    = dummyslices + [1:120:9961]; % e.g. 84 trials, 1 trial per
120 slices
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
config_serial(port,14400)         % PRISMA scanners only ; QUATTRO
& PENTA
start_serial                        % or start_cogent on Windows 7 /
Cogent PCs

for trail = 1:length(slicelist)
    % prepare each trial's stimulus here

    % wait for slice (& record which slice was found and when)
    [s(trial), t(trial)] = waitslice(port,slicelist(trial));
    % present the stimulus here

end

stop_serial                        % or stop_cogent on Windows 7 /
Cogent PCs
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

3T PRISMA and 7T MRI systems have additional information in the triggers. This can be accessed by the following extended syntax:

```
[codes, slices, times] = getslice( port ) % gets the slice type-code as well
as slice number and timestamps
[c,s,t] = waitslice(1,0,'V')             % waits for MRI sequence 'V'ersion
string on COM port 1
[c,s,t] = waitslice(1,0,'N')             % waits for the 'N'oise scan on
COM port 1
[c,s,t] = waitslice(1,12,'D')            % waits for 'D'ummy scan 12 on COM
port 1
[c,s,t] = waitslice(1,14,'R')            % waits for 'R'eference scan 14 on
COM port 1
[c,s,t] = waitslice(1,12,'F')            % waits for 'F'iller scan 12 on
COM port 1
```

```
[c,s,t] = waitslice(1,6,'B')           % waits for 'B' reverse blip scan  
6 on COM port 1  
[c,s,t] = waitslice(1,125,'S')         % waits for 'S'lice 125 -  
equivalent to waitslice(1,125)
```

Note that 'D'ummy & 'F'iller scan numbers count **downwards**, to 1.
'R'eference, 'B'lip & 'S'lice numbers count **up**, from 1.

NB Existing scripts will still work as before (as long as the serial port is configured as `config_serial(port,14400)`).

It is best to use slice triggers (`waitslice()`) to control timing between trails, and the normal `wait()` or `waituntil()` commands to control timing within a trial. An MRI scanner emulator is available to help with testing your script. You can install it, using Subversion/TortoiseSVN from here (it contains a demo script):

```
svn://bread.fil.ion.ucl.ac.uk/tools/cogent/Cogent2000extras/trunk/emulscan
```

Connection Diagram

From:

<http://intranet.fil.ion.ucl.ac.uk/wiki/cogentwiki/> - Cognitive Interface Wiki

Permanent link:

http://intranet.fil.ion.ucl.ac.uk/wiki/cogentwiki/doku.php?id=mri_scan_synchronisation

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